Project Title
(Recipe Finder)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Finding suitable recipes can be difficult when users search based on available ingredients rather than the exact recipe name. Traditional recipe-searching methods may require users to check many recipes individually, making the process time-consuming and inefficient. Therefore, there is a need for an efficient recipe finder that can quickly search, match, and retrieve recipes according to user requirements.

The proposed Recipe Finder is a Java-based application that allows users to search for recipes using recipe names, available ingredients, cuisine, and category. The system provides complete recipe information, including ingredients and preparation steps, and can support spelling correction and recipe selection based on factors such as cooking time, budget, rating, and nutritional values. Recipe information is organized in a structured text file, allowing the system to read, process, search, and display recipe records efficiently.

The project is implemented using Java in Visual Studio Code with a console-based user interface and text-file data storage. DSA techniques include Knuth–Morris–Pratt (KMP), Rabin–Karp, Edit Distance using Dynamic Programming, Bipartite Matching, Knapsack Optimization, and Randomized Hashing. These techniques are intended to support efficient pattern matching, ingredient matching, spelling correction, recipe selection, meal optimization, and fast recipe retrieval, demonstrating the practical application of DSA concepts in a real-world recipe search system

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